

Graphing Quadratics from Vertex or Standard Form

Answer Key

Algebra 1

Quick Answers

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|---|--|
| 1. (a) x-intercepts = -1, 5, (b) y-intercept = -5 | 7. (b) y-intercept = 7, — |
| 2. (a) x-intercepts = -3, 1, — | 8. (b) x-intercepts = -3, 3, — |
| 3. (a) x-intercepts = -1, 3, (b) y-intercept = -6 | 9. (b) x-intercepts = -1, 3, — |
| 4. (a) axis of symmetry = 3, (b) y-value of the vertex = -4 | 10. (a) x-intercepts = -1, 3, — |
| 5. (a) x-intercepts = -5, 1, (b) axis of symmetry = -2 | 11. (a) x-intercepts = -1, 3, (b) y-intercept = 3, — |
| 6. (a) x-intercepts = 1, 3, (b) y-value of the vertex = -2 | 12. (a) axis of symmetry = 4, (b) y-value of the vertex = -4, (c) x-intercepts = 2, 6, — |
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What is verified

5 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 2, 7, 8, 9, 10, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSF-BF.B.3 – *problems 7, 8, 9*

HSF-IF.C – *problems 1, 2, 3, 4, 5, 6, 10, 11, 12*

Difficulty 1–5 of 5 across 27 tagged checks.

1. $y = (x - 2)^2 - 9$. (a) x -intercepts. (b) y -intercept.

(a) The squared bracket is alone once 9 is added to both sides, so take square roots: $x - 2 = \pm 3$, giving $x = 5$ and $x = -1$.

$$\boxed{x = -1 \quad \text{and} \quad x = 5}$$

- (b) The y -intercept is the value at $x = 0$: $(0 - 2)^2 - 9 = 4 - 9$.

$$\boxed{-5}$$

2. $y = -(x + 1)^2 + 4$. (a) x -intercepts. (b) Opens up or down; maximum or minimum?

(a) $-(x + 1)^2 + 4 = 0$ gives $(x + 1)^2 = 4$, so $x + 1 = \pm 2$ and $x = 1$ or $x = -3$.

$$\boxed{x = -3 \quad \text{and} \quad x = 1}$$

(b) — **instructor-judged.** The parabola opens *downward*, so the vertex $(-1, 4)$ is a *maximum*. Full credit requires the reason read off the equation: the number multiplying the squared bracket is -1 , and a negative multiplier flips the parent graph over. A correct direction with no reason, or a reason taken from a sketch rather than the equation, is partial credit.

3. $y = 2(x - 1)^2 - 8$. (a) x -intercepts. (b) y -intercept.

(a) $2(x - 1)^2 = 8$, so $(x - 1)^2 = 4$ and $x - 1 = \pm 2$.

$$x = -1 \quad \text{and} \quad x = 3$$

(b) At $x = 0$: $2(0 - 1)^2 - 8 = 2 - 8$.

$$-6$$

4. $y = x^2 - 6x + 5$. (a) Axis of symmetry. (b) y -value of the vertex.

(a) Standard form with $a = 1$ and $b = -6$, so the axis is $x = -\frac{b}{2a} = -\frac{-6}{2}$.

$$x = 3$$

(b) The vertex sits on the axis, so substitute $x = 3$: $3^2 - 6(3) + 5 = 9 - 18 + 5$. The vertex is $(3, -4)$.

$$-4$$

5. $y = x^2 + 4x - 5$. (a) x -intercepts. (b) Axis of symmetry.

(a) Factor: product -5 , sum 4 gives $+5$ and -1 , so $(x + 5)(x - 1) = 0$.

$$x = -5 \quad \text{and} \quad x = 1$$

(b) $a = 1$, $b = 4$, so $x = -\frac{4}{2}$. Notice it is exactly halfway between the two intercepts.

$$x = -2$$

6. $y = 2x^2 - 8x + 6$. (a) x -intercepts. (b) y -value of the vertex.

(a) Take out the common factor 2 first: $2(x^2 - 4x + 3) = 2(x - 1)(x - 3) = 0$.

$$x = 1 \quad \text{and} \quad x = 3$$

(b) The axis is $x = -(-8)/(2 \cdot 2) = 2$, halfway between 1 and 3 . At $x = 2$: $2(4) - 16 + 6$.

$$-2$$

7. $y = x^2$ shifted 3 left and 2 down. (a) New equation in vertex form. (b) y -intercept.

(a) — **instructor-judged.** The equation must be $y = (x + 3)^2 - 2$. Full credit needs both signs right: a shift to the *left* puts a *plus* inside the bracket, and a shift down puts a minus outside. Writing $(x - 3)^2 + 2$ reverses both and earns no credit for (a), though (b) may still be marked against the student's own equation.

(b) At $x = 0$: $(0 + 3)^2 - 2 = 9 - 2$.

$$7$$

8. $y = x^2$ reflected in the x -axis, then shifted 9 up. (a) New equation. (b) x -intercepts.

(a) — **instructor-judged.** The equation must be $y = -x^2 + 9$. The reflection puts the minus in front of the squared term only; the shift then adds 9 to the whole thing. $y = -(x^2 + 9)$ is a different graph — it reflects the shifted parabola instead — and is not full credit.

(b) $-x^2 + 9 = 0$ gives $x^2 = 9$.

$$x = -3 \quad \text{and} \quad x = 3$$

9. $y = -2(x-1)^2 + 8$. (a) The three transformations from $y = x^2$, in order. (b) x -intercepts.

(a) — **instructor-judged**. Three moves, and all three are needed for full credit: shift 1 unit to the *right* (the bracket is $x - 1$); stretch vertically by a factor of 2 and reflect in the x -axis (the -2 in front); then shift 8 units up. The horizontal shift belongs inside the bracket and is applied before the vertical move. Naming only the shifts, or reading $(x - 1)$ as a shift to the left, is not full credit.

(b) $-2(x - 1)^2 + 8 = 0$ gives $(x - 1)^2 = 4$, so $x - 1 = \pm 2$.

$$x = -1 \quad \text{and} \quad x = 3$$

10. $y = (x - 1)^2 - 4$. (a) x -intercepts. (b) Sketch on the grid.

(a) $(x - 1)^2 = 4$, so $x - 1 = \pm 2$.

$$x = -1 \quad \text{and} \quad x = 3$$

(b) — **instructor-judged**. The sketch must open upward, with the lowest point plotted at $(1, -4)$ — read straight off the vertex form — and cross the horizontal axis at -1 and 3 . It must be symmetric about the vertical line through the lowest point. A curve through the two crossings whose lowest point is somewhere else, or a V-shape with a corner, is not full credit.

11. $y = -x^2 + 2x + 3$. (a) x -intercepts. (b) y -intercept. (c) Sketch on the grid.

(a) Multiply by -1 : $x^2 - 2x - 3 = 0$, so $(x - 3)(x + 1) = 0$.

$$x = -1 \quad \text{and} \quad x = 3$$

(b) At $x = 0$ the value is the constant term.

$$3$$

(c) — **instructor-judged**. The sketch must open downward (the coefficient of x^2 is negative), cross the horizontal axis at -1 and 3 , cross the vertical axis at 3 , and have its highest point plotted at $(1, 4)$ — the axis of symmetry is midway between the two crossings, and $-(1)^2 + 2(1) + 3 = 4$. An upward-opening curve, or a peak that is not on that middle line, is not full credit.

12. **Synthesis challenge.** $y = x^2 - 8x + 12$. (a) Axis of symmetry. (b) y -value of the vertex. (c) x -intercepts. (d) Sketch, and explain the symmetry check.

(a) $a = 1$, $b = -8$, so $x = -\frac{-8}{2}$.

$$x = 4$$

(b) At $x = 4$: $16 - 32 + 12$, so the vertex is $(4, -4)$.

$$-4$$

(c) Factor: product 12 , sum -8 gives -2 and -6 , so $(x - 2)(x - 6) = 0$.

$$x = 2 \quad \text{and} \quad x = 6$$

(d) — **instructor-judged**. The sketch must open upward, sit lowest at $(4, -4)$, and cross the horizontal axis at 2 and 6 . The written check must use symmetry: the two crossings are equally far from the line of symmetry — 2 is two units to its left, 6 is two units to its right — so their average has to be 4 . A student who sketches correctly but writes no symmetry argument has answered only half of (d).